

Dynamic Stochastic General Equilibrium and Business Cycles

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CHAPTER 2

Estimation of the Neo-Classical Growth Model

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HANDOUT OBJECTIVES

- Estimating the neo-classical growth model through likelihood methods;
- Incorporating trends and estimating them;
- Estimating multiple shocks;
- Ranking estimated models using a log-likelihood criterion;

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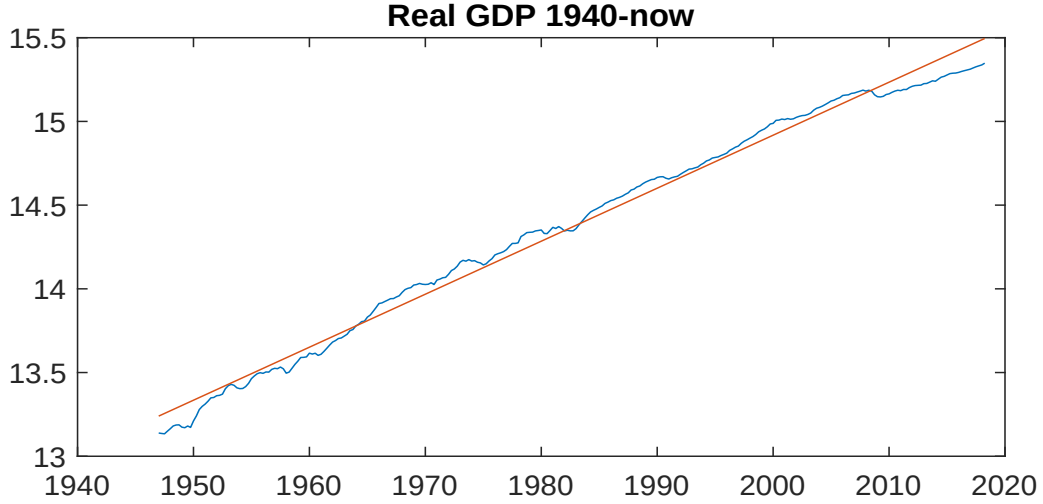


Figure 1: US per capita real GDP between 1945 to now

1. Estimation of the RBC model

Real business cycle theory (RBC theory) is a class of DSGE models in which business cycle fluctuations to a large extent can be accounted for by real shocks. In comparison/opposition, New Keynesian economics aims at explaining business cycles fluctuations by both nominal and real shocks. In RBC theory, prices are perfectly adjusting while governments and monetary policy are ineffective in smoothing short term fluctuations and should therefore concentrate on long-run structural policy.² In this section, we concentrate on the way the real business cycle model explains macroeconomic fluctuations through productivity shocks for the US economy. Using likelihood methods, we estimate US economy GDP deviations from its linear trend.

1.1 Detrending the RBC model

Recall our very simple RBC model with logarithmic utility, capital accumulation and Cobb-Douglas technology. The optimization problem is:

$$\begin{aligned} \max_{C_{jt}, K_{jt}} \mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^\tau \log(C_{jt+\tau}) \\ s.t. : C_{jt} + K_{jt} = A_t K_{jt-1}^\alpha + (1 - \delta) K_{jt-1} \end{aligned}$$

In the model above, there is no trend growth, the economy is assumed to converge to a fixed steady state value. In real life, the GDP of Western economies have steadily increased around an upward looking linear trend. See for instance [Figure 1](#) for the per capita real US GDP between 1947 and 2015.

In that set-up, labour is assumed to be fixed and inelastic. To introduce a trend, we need to have labour, even if it is exogenously fixed. The implicit production with fixed labor is:

$$Y_t = A_t K_{t-1}^\alpha \bar{H}^{1-\alpha}$$

²Policy interventions are justified by the presence of unemployment and inflation. RBC theory does not include these market imperfections contrary to New Keynesian economics.

where $\bar{H}$ is the fixed amount of hours worked involved in the production of new goods, it was previously assumed that $\bar{H} = 1$.³

Letting γ^t denotes a trend component to productivity. The economy is assumed to increase by 100 $(\gamma - 1)$ % each period. For instance with $\gamma = 1.03$, the GDP is assumed to increase steadily by 3% each period. This growth rate is distinct (and uncorrelated with) Total Factor Productivity (TFP) shock A_t . As in Solow's model or [Smets and Wouters \(2007\)](#), we assume technology is labour-augmenting, i.e. the production function is affected by a labour-augmenting trend, it reads as follows:

$$\begin{aligned} Y_t &= A_t K_{t-1}^\alpha \left(\bar{H} \gamma^t \right)^{1-\alpha} \\ &= A_t K_{t-1}^\alpha \gamma^{t(1-\alpha)} \end{aligned}$$

Given that trend in the technology of firms, the maximisation problem becomes:

$$\begin{aligned} \max_{C_{jt}, K_{jt}} \mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^\tau \log(C_{jt+\tau}) \\ s.t. : C_{jt+\tau} + K_{jt+\tau} = A_{t+\tau} K_{t-1+\tau}^\alpha \gamma^{(t+\tau)(1-\alpha)} + (1-\delta) K_{jt-1+\tau} \end{aligned}$$

The Euler condition now reads as:

$$\frac{1}{C_t} = \mathbb{E}_t \frac{1}{C_{t+1}} \beta \left[\alpha A_{t+1} K_t^{\alpha-1} \gamma^{(t+1)(1-\alpha)} + (1-\delta) \right] \quad (1)$$

Our model is characterized by 3 variables, their detrend counterpart is:

$$\begin{aligned} c_t &= C_t / \gamma^t \\ k_{t-1} &= K_{t-1} / \gamma^t \\ a_t &= A_t \end{aligned}$$

where c_t , k_{t-1} and a_t are stationary and per units of effective labor. Recall that K_{t-1} is a stock variable, the stock of capital in t is decided at the end of period $t-1$, its detrend counterpart thus requires to be detrended when the decision is taken, leading $k_{t-1} = K_{t-1} / \gamma^t$. We must detrend the two other equations. We don't need to transform A_t since it is exogenously stationary. Starting with the Euler equation:

$$\begin{aligned} \frac{1}{C_t} \frac{\gamma^t}{\gamma^t} &= \mathbb{E}_t \frac{1}{C_{t+1}} \frac{\gamma^{t+1}}{\gamma^{t+1}} \beta \left[\alpha A_{t+1} \left(K_t \frac{\gamma^{t+1}}{\gamma^{t+1}} \right)^{\alpha-1} \gamma^{(t+1)(1-\alpha)} + (1-\delta) \right] \\ \frac{1}{c_t} \frac{1}{\gamma^t} &= \mathbb{E}_t \frac{1}{c_{t+1}} \frac{1}{\gamma^{t+1}} \beta \left[\alpha A_{t+1} \left(k_t \gamma^{t+1} \right)^{\alpha-1} \gamma^{(t+1)(1-\alpha)} + (1-\delta) \right] \\ \frac{1}{c_t} &= \mathbb{E}_t \frac{1}{c_{t+1}} \frac{\gamma^t}{\gamma^{t+1}} \beta \left[\alpha A_{t+1} (k_t)^{\alpha-1} + (1-\delta) \right] \\ \frac{1}{c_t} &= \mathbb{E}_t \frac{1}{c_{t+1}} \frac{\beta}{\gamma} \left[\alpha A_{t+1} (k_t)^{\alpha-1} + (1-\delta) \right] \end{aligned}$$

³In next lectures, we will have variable hours worked. By simplification and for tractability reasons, we assume hours worked are fixed.

The budget constraint is:

$$\begin{aligned}
 C_t + K_t &= A_t K_{t-1}^\alpha \gamma^{t(1-\alpha)} + (1 - \delta) K_{t-1} \\
 C_t \frac{\gamma^t}{\gamma^t} + K_t \frac{\gamma^{t+1}}{\gamma^{t+1}} &= A_t \left(K_{t-1} \frac{\gamma^t}{\gamma^t} \right)^\alpha \gamma^{t(1-\alpha)} + (1 - \delta) K_{t-1} \frac{\gamma^t}{\gamma^t} \\
 c_t \gamma^t + k_t \gamma^{t+1} &= A_t (k_{t-1} \gamma^t)^\alpha \gamma^{t(1-\alpha)} + (1 - \delta) k_{t-1} \gamma^t \\
 c_t + k_t \frac{\gamma^{t+1}}{\gamma^t} &= A_t k_{t-1}^\alpha + (1 - \delta) k_{t-1} \\
 c_t + \gamma k_t &= A_t k_{t-1}^\alpha + (1 - \delta) k_{t-1}
 \end{aligned}$$

Finally we add output in the model with $Y_t = A_t K_{t-1}^\alpha \gamma^{t(1-\alpha)}$. Detrending the equation is straightforward, $y_t = a_t k_{t-1}^\alpha$. Our detrended model is now:

$$\begin{aligned}
 \frac{1}{c_t} &= \mathbb{E}_t \frac{1}{c_{t+1}} \frac{\beta}{\gamma} \left[\alpha A_{t+1} k_t^{\alpha-1} + (1 - \delta) \right] \\
 c_t + \gamma k_t &= A_t k_{t-1}^\alpha + (1 - \delta) k_{t-1} \\
 y_t &= a_t k_{t-1}^\alpha \\
 \log(A_t) &= \rho \log(A_{t-1}) + \log(\eta_t)
 \end{aligned}$$

Letting $\bar{k}$ and $\bar{y}$ denote the detrended steady state of capital and production, the steady state reads as:

$$\begin{aligned}
 \bar{k} &= \left(\left[\frac{\gamma}{\beta} - (1 - \delta) \right] \frac{1}{\alpha} \right)^{1/(\alpha-1)} \\
 \bar{c} &= \bar{k}^\alpha + (1 - \delta - \gamma) \bar{k} \\
 \bar{y} &= \bar{k}^\alpha
 \end{aligned}$$

The model's steady state and dynamics are very slightly affected by the trend component.

Taken in logs, our linear system now reads as follows:

$$\hat{c}_t - \mathbb{E}_t \hat{c}_{t+1} = -\frac{\beta}{\gamma} \alpha \bar{k}^{\alpha-1} \left(\mathbb{E}_t \hat{a}_{t+1} - (1 - \alpha) \hat{k}_t \right) \quad (2)$$

$$\bar{c} \hat{c}_t + \gamma \bar{k} \hat{k}_t = \bar{k}^\alpha \left(\hat{a}_t + \alpha \hat{k}_{t-1} \right) + (1 - \delta) \bar{k} \hat{k}_{t-1} \quad (3)$$

$$\hat{y}_t = a_t + \alpha \hat{k}_{t-1} \quad (4)$$

$$\hat{a}_t = \rho \hat{a}_{t-1} + \eta_t \quad (5)$$

We are now ready to estimate the model which includes a trend component.

1.2 Data transformation and Trend Slope

A key problem when performing fit exercises with DSGE models is to map a stationary model with non stationary-data. Macroeconomic time series, like GDP, are better characterized as non-stationary processes that have no tendency to return to a deterministic path as underlined by [Nelson and Plosser \(1982\)](#) and thus requires data transformation to get them stationary.

There are two main components which makes the GDP steadily grow (as underlined by the [Solow \(1956\)](#) model):

1. Technological changes.

2. The population.

First, the first way to remove a non stationary component in the GDP is to divide it by the working population. Indeed, the growth of the population makes the GDP steadily increase at least at the same rate as the population. It is also possible to remove technological changes by rewriting the model in terms of effective labour units as in Solow's model (this step is explained in the following subsection).

Let's first consider the real GDP. You can find for free many macroeconomic time series using DB·NOMICS. It's easy to use (see the DB·NOMICS guide). Once you have downloaded the DB·NOMICS function and have it in your current folder, follow these steps:

1. go to <https://db.nomics.world/>
2. scroll down on the main page, find and click on the OECD database, select the quaterly national account, and find the token (i.e. the key of dbnomics). If you are lazy, the token is:

OECD/QNA/USA.B1_GS1.LNBQRSA.Q

3. Create a Matlab file called "mydb.m" and use the dbnomics function with the token to import the series into MATLAB.
4. Edit `mydata.m` and add the matrix tags for Matlab to read the data. It should look like the following code:

```
1 [my_mat,~,T] = call_dbnomics('OECD/QNA/USA.B1_GS1.LNBQRSA.Q');
```

5. Next, we declare the corresponding vector of quarters, denoted `T`, converting the dates in matlab format for plotting purposes. Here `gross_y` denotes the GDP before transformation (stored in the 2nd column of the data matrix).

```
1 gross_y = my_mat(:,2);
```

6. Recall that in the model, each random variable $\hat{x}_t$ is expressed in terms of detrended percentage deviations from the steady state:

$$\hat{x}_t \simeq \log \left(\frac{X_t}{\bar{X} \gamma^t} \right) \quad (6)$$

where X_t is the non-linear expression of $\hat{x}_t$, $\bar{X}$ is the empirical mean (or the steady state) of X_t and γ^t is the trend component. We transform our sample in the same vein, we first express in terms of log deviations from the trend $\hat{X}_t \simeq \log(X_t/\gamma^t)$, we store the result in a variable called `ln_gap` in Matlab. Then, we remove the steady state, i.e. the mean of the sample such that $\hat{x}_t = \hat{x}_t - \mu(\hat{X}_t)$ where $\mu(\cdot)$ is the empirical mean, and add the resulting values to the chart.

To have our in data consistent with the model, (a) we need to apply logs to our data, (b) remove the trend γ^t , (c) remove the mean/steady state:

- a First , we apply the log. Variable `ln_y` is the log of output:

```
1 ln_y = log(gross_y);
```

- b Second we want to remove trend. To do so, we want to know the value of the trend γ that has to be calibrated in the model before the estimation. This parameter is the slope of the trend. Simple linear regression considers only one independent variable using the relation:

$$y_t = \beta_0 + \beta_1 x_t + \epsilon_t$$

where $t = [1, 2, \dots, T]$ is the number of period, y_t is the value of log GDP and ϵ_t is the log deviation from the trend. Here, the trend component is $\beta_1 = \gamma$. We perform the estimation of the trend in Matlab through:

```
1 X = [ones(length(ln_y),1) (1:length(ln_y))'];
2 OLS= X\ln_y;
3 trend_y = X*OLS;
4 gamma = 1+OLS(2);
```

The estimated value of $\gamma = 1.005$, i.e. the US economy has experienced a 0.5% quarterly growth rate since 1960. Variable `ln_gap` denotes the detrended output. However, the empirical mean of the output is (still) different from 0.

- c Finally, we remove the empirical mean (which can be interpreted as the steady state of the model):

```
1 ln_gap = ln_y-trend_y;
2 y = ln_gap-mean(ln_gap);
```

We can plot our result:

```
1 figure;
2 plot(T,ln_y,T,trend_y)
3 title('Real GDP 1940-now')
4 figure;
5 plot(T,y)
6 title('Real GDP variations around the trend')
```

And save the transformed series of output into a matrix file, called `mydata.mat`, which will be read by Dynare right after for the fit exercise.

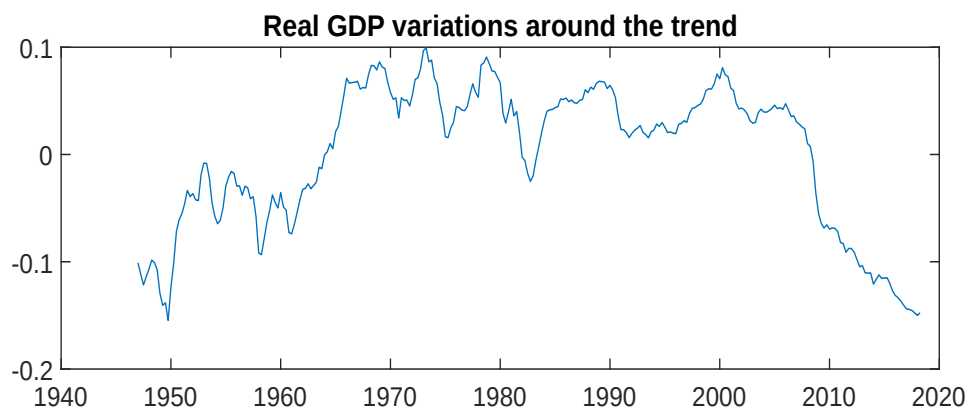


Figure 2: Percentage deviation of the US GDP from its linear trend

```
1 save mydata.mat y;
```

Figure 2 plots the transformed GDP, which can be interpreted as a measure of the detrended output gap. It is thus possible to explain the gap between the real GDP and its linear trend through productivity shocks.

Important note: as we are going to call via Dynare the Matlab file "[mydb.m](#)" we've just created, variable `y` should be declared as an endogenous variable to be estimated in the Dynare file.

1.3 Dynare syntax for log-likelihood estimation

First, let us take back the file from the first handout. Then:

1. Include output equation and variable.
2. Create a new trend variable γ and calibrate it such that $\gamma = 1.005$.
3. Replace the old steady state with the new one, same for linear equations.

Second, to estimate using maximum likelihood techniques, first declare the variables of the model which are observed. This is done using the `varobs VAR1 VAR2 VAR3 ;` command, `VARX` must be in the data file a vector of an observed variable, otherwise the estimation won't start. Here, we only have output as observed:

code

```
varobs y;
```

Next, we specify the parameters to be estimated in the `estimated_params;` block. When writing the estimated params section, each parameter to be estimated needs to be given an initial value, and can be given bounds: the syntax to do this is `parameter, initial value [ lower bound, upper bound ]`; where the parts in the square brackets are optional. For the standard errors of the shocks, we use the same syntax, preceding the exogenous variable with `stderr`, like for specifying the shocks. The full estimated params section is:

code

```
estimated_params;  
    stderr    e_a,    0.007,    0,    inf;  
            rho,    0.95,    0,    1;  
end;
```

Finally, we need to call estimation and specify where Dynare can find the observed series. Our observed vector is in the Matlab file "[mydata.mat](#)". This is called with:

code

```
estimation(datafile=mydata);
```

1.4 Results from the estimation exercise

Dynare is now solving the optimization problem by finding the value of ρ and the sequence of productivity shocks $\{\eta_t\}_{t=1}^T$ (where T is the size of the sample) which maximize the log-likelihood function. At the beginning of the optimization, the value of the log-likelihood function is [-9325.082](#), while at the end, it's value is [-762.489409](#).

console response

```

Loading 222 observations from mydata.m
Initial value of the log posterior (or likelihood): -9325.082
_____

f at the beginning of new iteration, 9325.0819914307
Predicted improvement: 458267455.582368310
lambda = 1; f = 9468.9528837
lambda = 0.33333; f = 9340.8022101
lambda = 0.11111; f = 9326.7416852
lambda = 0.037037; f = 9325.2388843
lambda = 0.012346; f = 9325.0917329
lambda = 0.0041152; f = -126.8987819
lambda = 0.0013717; f = -439.0378258
lambda = 0.00045725; f = -671.6612365
lambda = 0.00015242; f = -741.4731680
lambda = 5.0805e-05; f = -76.4714181
lambda = 1.6935e-05; f = 2305.8927345
Norm of dx 302.74
_____

Improvement on iteration 1 = 10066.555159427
_____

.....

Improvement on iteration 19 = 0.000000000
improvement < crit termination
Objective function at mode: -762.489409
RESULTS FROM MAXIMUM LIKELIHOOD ESTIMATION
parameters
  Estimate s.d. t-stat
rho 0.9818 0.0125 78.5833
standard deviation of shocks
  Estimate s.d. t-stat
e_a 0.0082 0.0004 21.2207
Total computing time : 0h00m08s

```

According to the estimation output, the optimized parameters are:

$$\rho = 0.9861$$

$$sd(\eta_t) = 0.0093$$

Dynare also gives the standard deviation (the lower is the standard deviation, the better is the identification of the parameter) and the t-stats. Any parameter with the t-stat above 1.96 is statistically significant at the 95% level.

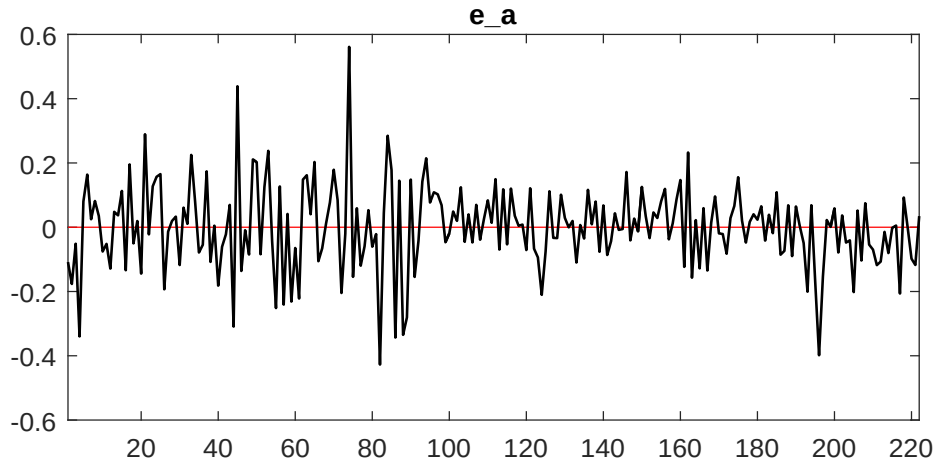


Figure 3: The sequence of estimated smoothed shocks $\{\eta_t\}_{t=1}^T$ over the sample period

Figure 3 plots the estimated sequence of the productivity shock which replicates the best the observed GDP over the sample period.

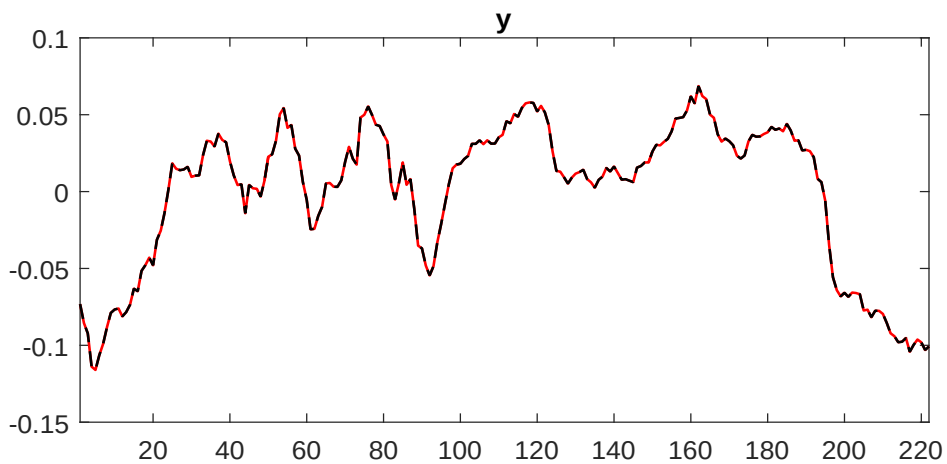


Figure 4: Observed series (black) and model-generated series (red) which use the estimated sequence of the productivity shock drawn in Figure 3

Finally, we can compare the observed versus the model-generated series in Figure 4. Our model is fairly able to replicate the business cycles of the US economy over the sample period.

The estimation results are saved in the `oo_` structure in the workspace. Some details are given in Table 1.

Stored in	Description
oo_.SmoothedVariables	Estimated values of each endogenous variable for all time periods.
oo_.SmoothedShocks	Estimated values of each shock for all time periods.

Table 1: Matlab's workspace after a likelihood estimation

It is possible to reverse the detrending method in order to get estimated data back to their raw values. This operation may look useless at first sight, but it allows us to plot and analyse the unobserved path of capital and productivity over the business cycles. First, we can plot detrended series:

code

```
load mydata;
figure
subplot(3,1,1);
plot(T,oo_.SmoothedVariables.y)
title('Output')
subplot(3,1,2);
plot(T,oo_.SmoothedVariables.k)
title('Capital')
subplot(3,1,3);
plot(T,oo_.SmoothedVariables.a)
title('Productivity')
```

The result from the previous code is drawn in [Figure 5](#). From this figure, it is possible to explain the persistence of negative output gaps since the financial crisis by an important decline in physical capital stock. Another important feature of the business cycles is the variance of productivity shocks which dropped strongly after 1980 (by the way do you know what have made the business cycles different after 1980?)

To reverse the detrending method, we just need to add up the trend component and apply the exponential function `exp(·)` to reverse the log. We don't need to plot the productivity shock as it is assumed to be detrended and stationary.

code

```
% REVERSING
figure;
subplot(2,1,1);
plot(T,exp(trend_y+oo_.SmoothedVariables.y))
title('Output')
subplot(2,1,2);
plot(T,exp(trend_y+oo_.SmoothedVariables.k))
title('Capital')
```

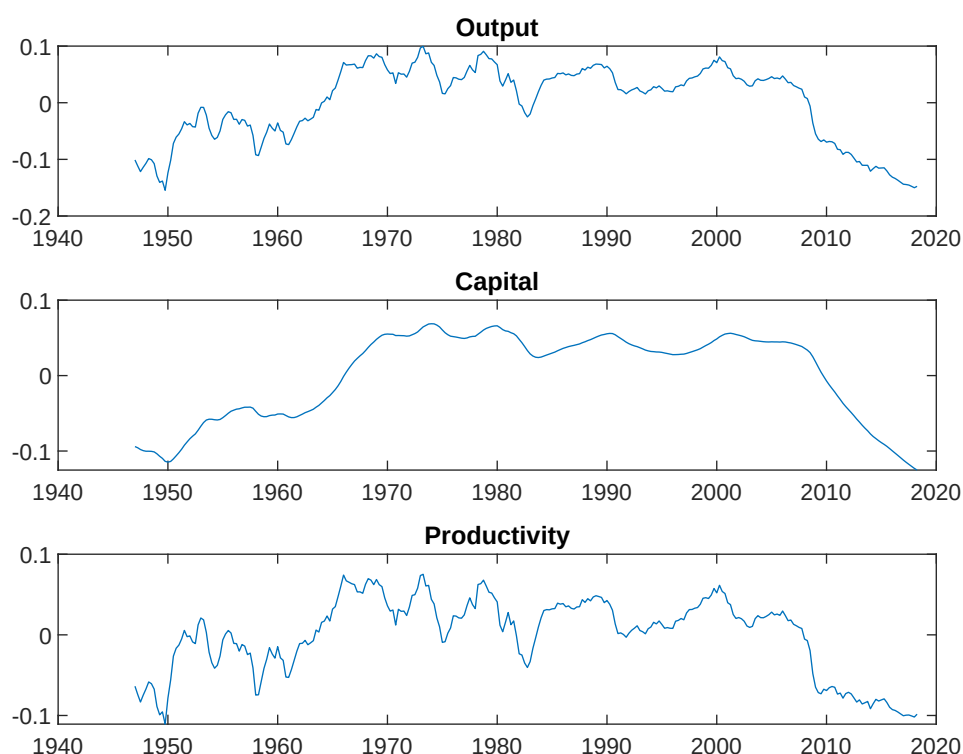


Figure 5: Model-generated output, capital and productivity (using the estimated model and sequence of shocks).

Exercise 1: Second moment statistics and IRFs of the estimated model

1. Compute using Matlab's function `std(.)` the standard deviation of the estimated sequence of the productivity shock. Do you find the same result as Dynare?
2. Compare the standard deviation of the model with the one observed in the data. Is your model relevant ?
3. Same question with the autocorrelation up to five order. Is your model relevant?
4. In your opinion, why is it important to have second order statistics relevant with the data?
5. Plot the IRFs of the estimated model.

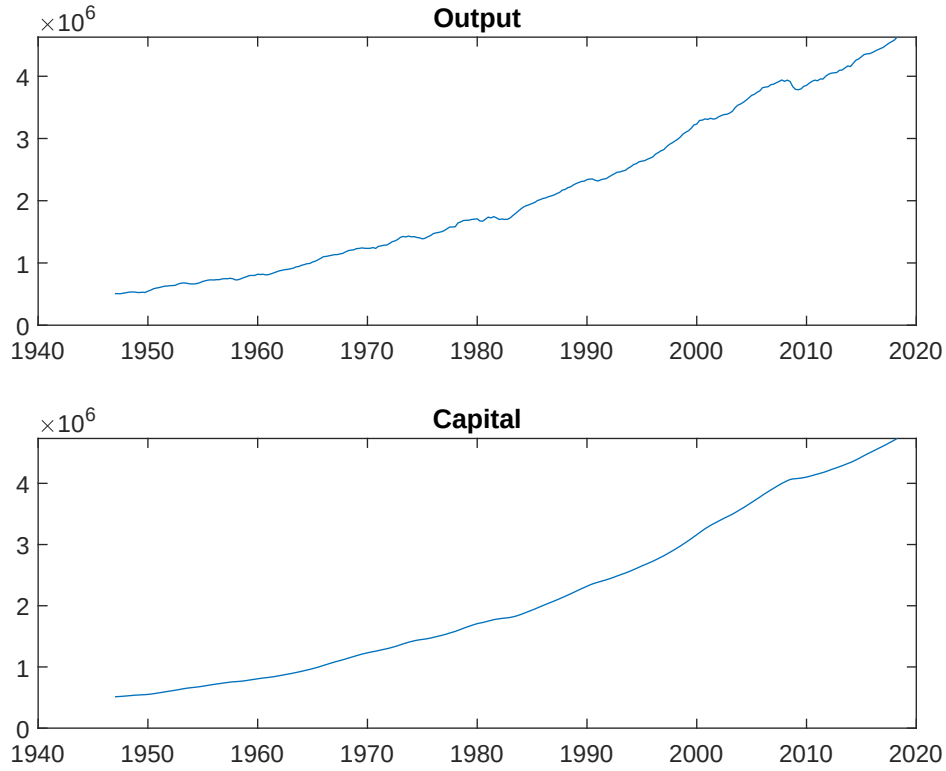


Figure 6: GDP and physical capital generated by the model after reversing the detrending method

Exercise 2: Estimating the model with output and consumption

Consider now adding another time serie in the model: consumption. It is thus possible to get directly from dbnomics.

To avoid stochastic singularity issue, you must have the same number of shocks (or more) than observables. Estimation is ok as long as estimated parameters (including shock variances) are identified (i.e. different from 0 and not estimated on their lower/upper bound). To estimate consumption, we need to add one more shock to the model. To increase demand shock effects, we also include investment I_t in the GDP.

Let's consider a public spending shock in the GDP:

$$Y_t = C_t + I_t + G_t$$

Public spending G_t is financed by a tax T_{jt} charged to all households populating our economy, assuming a balanced government budget over the cycles:

$$G_t = \int_0^1 T_{jt} dj$$

This tax affects the budget constraint of the household:

$$C_{jt} + K_{jt} + T_{jt} = A_t K_{t-1}^\alpha \gamma^{(t+\tau)(1-\alpha)} + (1 - \delta) K_{jt-1}$$

Spending are exogenously determined as a constant fraction of the GDP ($\bar{G} = g^y \bar{Y}$):

$$G_t = \bar{G} \varepsilon_t^G \text{ with } \log(\varepsilon_t^G) = \rho_G \log(\varepsilon_{t-1}^G) + \eta_t^G$$

To summarize with the detrended spending variable $g_t = G_t/\gamma^t$, our detrended model is now summarized by:

$$\hat{c}_t = -\frac{\beta}{\gamma} \alpha \bar{k}^{\alpha-1} (\mathbb{E}_t \hat{a}_{t+1} - (1-\alpha) \hat{k}_t) + \beta [\alpha \bar{k}^{\alpha-1} + (1-\delta)] \mathbb{E}_t \hat{c}_{t+1} \quad (7)$$

$$\hat{y}_t = \hat{a}_t + \alpha \hat{k}_{t-1} \quad (8)$$

$$\bar{y} \hat{y}_t = \bar{c} \hat{c}_t + \bar{u}_t + g^y \bar{y} \hat{g}_t \quad (9)$$

$$\bar{u}_t = \gamma \bar{k} \hat{k}_t - (1-\delta) \bar{k} \hat{k}_{t-1} \quad (10)$$

$$\hat{a}_t = \rho_A \hat{a}_{t-1} + \eta_t^A \quad (11)$$

$$\hat{g}_t = \rho_G \hat{g}_{t-1} + \eta_t^G \quad (12)$$

with $g^y = 0.2$, *i.e.* spending shocks accounts for 20% of output variations. The steady state thus reads as follows:

$$\begin{aligned} \bar{k} &= \left(\left[\frac{\gamma}{\beta} - (1-\delta) \right] \frac{1}{\alpha} \right)^{1/(\alpha-1)} \\ \bar{i} &= (\gamma - 1 + \delta) \bar{k} \\ \bar{y} &= \bar{k}^\alpha \\ \bar{c} &= (1 - g^y) \bar{y} - \bar{i} \end{aligned}$$

Our economy is characterized by a set of six variables $\{\hat{c}_t, \hat{k}_t, \hat{y}_t, \hat{u}_t, \hat{a}_t, \hat{g}_t\}$.

1. Code the new model.
2. Estimate the new model with consumption and output, estimate the variance of the spending shock as well as its AR(1) term.
3. Analyse the variance decomposition. Which shock is driving/explaining the business cycles for the US economy?
4. Is the slope of the trend different for consumption and the GDP? (In the next exercise, we will develop a method to estimate one single trend for both consumption and GDP).
5. Let's try to develop and compare an alternative shock which could explain the US business cycles. Include the following preference shock to the model:

$$\max_{C_{jt}, K_{jt}} \mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^\tau \varepsilon_{t+\tau}^C \log(C_{jt+\tau})$$

- (a) How the Euler equation is affected by this shock?
- (b) Estimate this new model with its AR(1) terms defined as:

$$\log(\varepsilon_t^C) = \rho_C \log(\varepsilon_{t-1}^C) + \eta_t^C$$

- (c) Which model is better given the data? (compare the model with government spending shock, preference shock and both shocks).

Exercise 3: Estimating deep parameters

Consider in this exercise the simple model without consumption/investment. It is possible to estimate parameters such as β , α and δ . However, these parameters affect the steady state of the model. Indeed, recall that:

$$\begin{aligned}\bar{k} &= \left(\left[\frac{\gamma}{\beta} - (1 - \delta) \right] \frac{1}{\alpha} \right)^{1/(\alpha-1)} \\ \bar{c} &= \bar{k}^\alpha + (1 - \delta - \gamma) \bar{k} \\ \bar{y} &= \bar{k}^\alpha\end{aligned}$$

Each time β , α and δ are changed to maximize the log-likelihood function, the steady state should be updated accordingly. It is thus easy in Dynare to update the steady of the model. In `model(linear);`'s block, add:

code

```
#Kss = ((gamma/beta-(1-delta))/alpha)^(1/(alpha-1));
#Css = Kss^alpha - (1-delta-gamma)*Kss;
```

At the beginning of your Dynare file, you shall remove Kss and Css from the parameter's block `parameters ...`; to make dynare update these parameters each iteration.

Any parameter in model's block preceded by the sign "`#`" is updated by Dynare each iteration of the optimization algorithm employed in the fit exercise.

1. Estimate the value δ and look at the t-stat. Is this parameter statistically significant? From this result, try to explain why δ is not well identified.
2. Do you find a similar result for β , α ? Given the data, is your fit better with or without these parameters?
3. [Smets and Wouters \(2007\)](#) obtain a well identified estimation of α and β for the US economy. Do you find such a result with that simple RBC model?

Exercise 4: Estimating trends

It is also possible to estimate directly the value of the trend γ as in [Smets and Wouters \(2007\)](#). The basic idea of this approach is to map the model to the first difference data. This approach is interesting as first difference filtering *should* remove unit root component in the data, and allows us to estimate the trend component rather than calibrating it.

Let's assume $[y_t^{obs}, c_t^{obs}]$ is the matrix of observable variables (*i.e.* GDP) and $[\hat{y}_t, \hat{c}_t]$ are their theoretical (and linear) counterpart in the model. Assuming $[y_t^{obs}, c_t^{obs}]$ are respectively the growth rates of the GDP and consumption, their corresponding counterpart in the model are:

$$\begin{aligned}y_t^{obs} &= \log(\gamma) + \hat{y}_t - \hat{y}_{t-1} \\c_t^{obs} &= \log(\gamma) + \hat{c}_t - \hat{c}_{t-1}\end{aligned}$$

where $\log(\gamma)$ is the common growth rate/trend to consumption and output.

1. Create two new endogenous variables in the model, called `y_obs` and `c_obs`, and translate the two previous mentioned equations into your Dynare code.
2. Update the `varobs y;` tag.
3. Transform your data applying a first difference filtering. For instance for the GDP:

code

```
% taking in logs GDP
ln_y = log(gross_y);
% growth rate of GDP
y_obs = diff(ln_y);
```

4. The steady state of the model is affected by γ , since we plan to estimate γ and relying on the previous exercise, edit your code to enable the steady state update during the fit exercise. Is this parameter well identified? Is this parameter above of below the value of γ we computed before?

References

- Nelson, C. R., Plosser, C. R., 1982. Trends and random walks in macroeconomic time series: some evidence and implications. *Journal of monetary economics* 10 (2), 139–162.
- Smets, F., Wouters, R., 2007. Shocks and frictions in us business cycles: A bayesian dsge approach. National bank of belgium working paper (109).
- Solow, R. M., 1956. A contribution to the theory of economic growth. *The quarterly journal of economics*, 65–94.